Ternary electrolyte: $\frac{N_1\delta}{D_1 c_{1,bulk}} = -\frac{2-\gamma+3\delta^{2/3}}{1-\delta}$, $\gamma = \frac{-z_3 c_{3,bulk}}{z_2 c_{2,bulk}}$

↳ $\gamma=0 \rightarrow$ binary electrolyte $\Rightarrow \frac{N_1\delta}{D_1 c_{1,bulk}} = -2$

↳ $\gamma=1 \rightarrow$ XS supporting electrolyte $\Rightarrow \frac{N_1\delta}{D_1 c_{1,bulk}} = -1$

(when $t_j \to 0$, we neglect migration)

Limiting cases: 1) Uniform composition $\frac{i_{avg}}{i_{lim}} << 1$ ← corresponds maximum change in conc. in the BL

$\nabla^2\phi = 0 \Rightarrow \phi = \frac{i_{avg}}{\kappa}(L-x)$

→ Broke this down into charge transfer + ohmic resistances

$R_{ct} = \left.\frac{\partial(V-\phi)}{\partial i}\right|_{i_{avg}}$

$R_\Omega = \left.\frac{\partial\phi}{\partial i}\right|_{i_{avg}}$

R_{ct}/R_Ω ratio would tell you what current regime you would in

↳ Limiting current density ($c_{i,lim} = 0$ @ surface): $\nabla^2 c_{i,lim} = 0 \Rightarrow c_{i,lim} = \frac{c_{i,bulk}}{\delta} x$

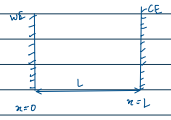
$i_{lim} = \frac{nF D_{i,lim}}{s_{i,lim}} \frac{c_{i,bulk}}{\delta}$

Using $R_{ct} \& R_\Omega$ (for uniform composition) → what cell length L, would cause current density to be "ohmically limited"

(Tafel kinetics) $\frac{R_{ct}}{R_\Omega} = \frac{RT}{\alpha_c F i_{avg}}\left(\frac{\kappa}{L}\right)$ If $\frac{R_{ct}}{R_\Omega} << 1 \rightarrow$ dominated by ohmic resistance

comes from differentiating B-V i expression

Assume $\kappa = 0.1\ S\ cm^{-1}$, $i_0 = 0.1\ mA\ cm^{-2}$, $\alpha_c = 0.5$, @ $i_{avg} = 100\ mA/cm^2$ } $L = 5\ cm$ ← characteristic cell length



If we instead assume linear kinetics, how does V and Q_o vary w/ $\frac{R_{ct}}{R_\Omega}$?

linear. $i_{avg} = \frac{i_0 F(\alpha_a+\alpha_c)}{RT}(V-\phi_o)$ ↳ potential at the electrode

$\phi_o = \frac{i_{avg} L}{\kappa}$

$\left.\frac{\partial(V-\phi)}{\partial i}\right|_{i_{avg}} = \frac{RT}{i_0 F(\alpha_a+\alpha_c)} = R_{ct}$

$\left.\frac{\partial\phi}{\partial i}\right|_{i_{avg}} = \frac{L}{\kappa} = R_\Omega$

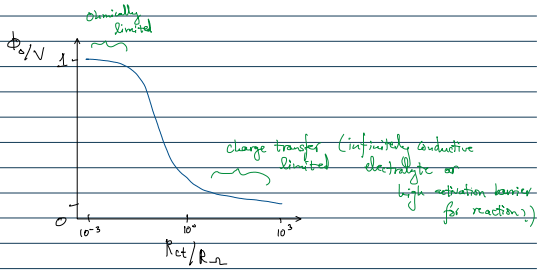
Independent of i_{avg}

$\frac{R_{ct}}{R_\Omega} = \frac{RT}{(i_0)F(\alpha_a+\alpha_c)} \cdot \left(\frac{\kappa}{L}\right)$

plugging into $\phi_o = i_{avg}\frac{L}{\kappa}$

$\phi_o = \frac{i_0(\alpha_a+\alpha_c)FL}{RT\,\kappa}(V-\phi_o)$

$\frac{\phi_o}{(V-\phi_o)} = \frac{i_0(\alpha_a+\alpha_c)FL}{RT\,\kappa} = \frac{R_\Omega}{R_{ct}}$



$$\frac{\phi_a}{V} = \frac{1}{w_a} \cdot \quad w_a = \frac{R-n}{R_{ct}}$$

Linear kinetics

Limiting Currents:

$\nabla^2 c_{i,lim} = 0$, BC's : @ $x=0$, $c_{i,lim} = 0$

@ $x=\delta$, $c_{i,lim} = c_{i,bulk}$

$$c_{i,lim} = \frac{c_{i,bulk}\, x}{\delta} \quad ; \quad i_{lim} = \frac{nF D_{i,lim} c_{i,bulk}}{\delta_{lim}\ \delta}$$

What will be i, if $c_{surf} \neq 0$, assuming $t_{i,lim} = 0$, i.e. diffusion only?

$\nabla^2 c = 0$, BC @ $x=0$, $c = c_{surf}$

@ $x=\delta$, $c = c_{bulk}$

$c = Ax + B$

$$\boxed{c = \frac{c_b - c_{surf}}{\delta} x + c_{surf}}$$

↳ Faraday's law: $i = \frac{nF D_j}{-s_j} \left.\frac{dc_j}{dx}\right|_{x=0} = \frac{nF D_j}{-s_j}\left(\frac{c_{bulk} - c_{surf}}{\delta}\right)$ ← If we measure i & then have δ (from BL analysis) ↳ then can find c_{surf} → Conc of H^+ will give local pH